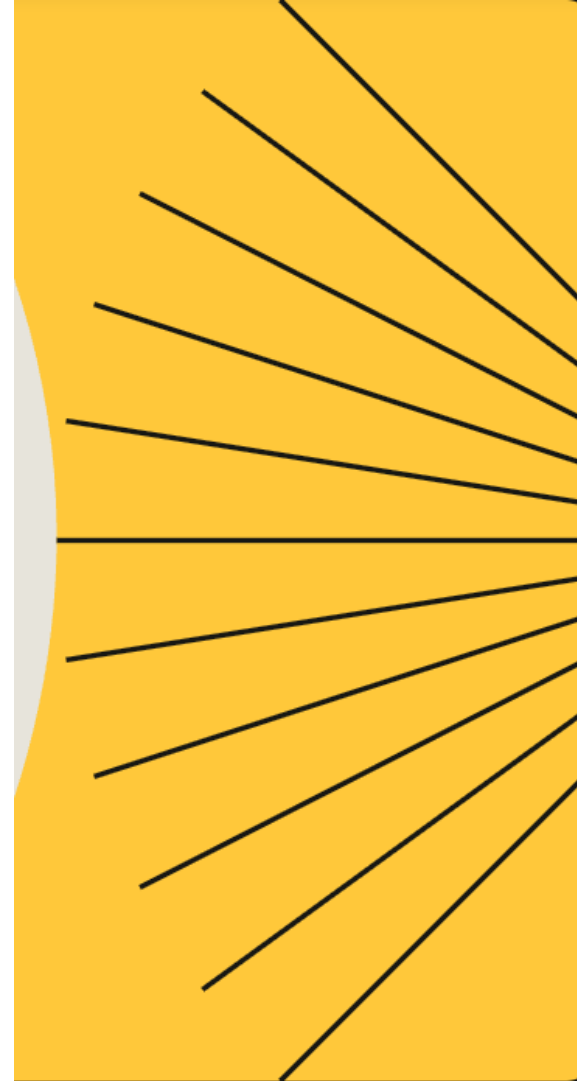


Clinical Pattern Recognition & Decision Support Platform

Saurav Upadhyaya



Healthcare Fraud: Two Schemes



The Limits of Traditional Fraud Detection

Manual & Rule-Based Systems

Labor-intensive, inflexible, and unable to adapt to sophisticated, evolving fraud tactics at the scale of millions of claims (*TASHIN et al., 2024; Pingili, 2025*).

Data Quality Barriers

Inconsistent standards, poor data quality, limited real-time detection, and scarce labeled datasets constrain supervised learning effectiveness (*Hamid et al., 2024; TASHIN et al., 2024*).

AI-Driven Frameworks: Progress and Gaps

What's working

Advanced AI-driven frameworks integrate diverse data sources, including structured claims data, patient demographics, provider information, external fraud alerts, and regulatory guidelines, with robust anonymization to ensure privacy (TASHIN et al., 2024; Pingili, 2025).

Critical Gaps

- High false-positive rates requiring human review
- Data privacy regulations (HIPAA, GDPR), inconsistent terminologies, algorithmic bias, model opacity,
- Integration challenges with legacy systems
- Necessity of ongoing human oversight (Hamid et al., 2024; TASHIN et al., 2024; Pingili, 2025).

Advanced Ingestion Pipeline & Clinical Context Engine

CMS DE-SynPUF

Processes wide-format outpatient data, normalizing Claim ID, Patient ID, Service Date, and Payment Amount fields.

Technology Stack

LangGraph, Llama 3.3 (70B), ChromaDB, FastAPI, Groq LLM, SQLite, and PII guardrails for secure, scalable processing.

Choreographer Timeline Pattern

Clusters claims into patient-day clinical windows, enabling context-aware, temporally grounded fraud analysis.

Dual-Core Compliance Matrix & Auditor Interface

Deterministic Gate

Applies **4,300+** CMS NCCI edits for instant detection of unauthorized procedural splits and unbundling violations.

Probabilistic Gate

Profiles billing outliers using Gaussian distribution and Z-score analysis to flag statistical anomalies.

Auditor Interface

- Web-based application with real-time flags, blocked overpayments
- NCCI violations Interactive review queues for upcoding and unbundling alerts
- Integrated AI-powered chatbot with text-to-SQL for rapid, natural-language auditor queries

Product Development

1. Data Ingestion

CMS DE-SynPUF, SQLite with PII guardrails.

2. Fraud Analytics

NCCI deterministic gate, probabilistic Z-score, choreographer timeline.

3. API Gateways

FastAPI, LangGraph agent, LLM endpoints.

4. Web Interface

Dashboard and AI-powered Chatbot

Product Development - Chat

1. SQL Generation

- 1) Allows: SELECT, WITH/CTE
- 2) Blocks: DROP, INSERT, UPDATE, SELECT *, PII column reference, multi-statement

2. Questions Not Answered

- 1) Give me the SSN for member M12..
- 2) Show me the email and phone number on claim..

3. PII

- 1) SSN fully redacted, email partially redacted, none redacted to last 4,
- 2) Non-PII column untouched

4. Sample Questions

- 1) Total paid amount, claims pending
- 2) DRG code with highest overpayment exposure

Metrics Evaluation - Chat

- What is the total paid amount across all claims?
loops=1 rows=1 latency=1.70s safe=True
context_precision=1.00 faithfulness=1.00 answer_relevance=1.00
- How many claims are pending?
loops=1 rows=1 latency=1.97s safe=True
context_precision=1.00 faithfulness=1.00 answer_relevance=1.00
- Which DRG code has the highest total overpayment exposure?
loops=1 rows=1 latency=1.28s safe=True
context_precision=1.00 faithfulness=1.00 answer_relevance=1.00
- List denied Medicare Advantage claims with their billed amounts
loops=1 rows=13 latency=1.68s safe=True
context_precision=1.00 faithfulness=1.00 answer_relevance=1.00
- What is the average days in hospital for FWA-flagged claims?
loops=1 rows=1 latency=12.09s safe=True
context_precision=1.00 faithfulness=1.00 answer_relevance=1.00
- Break down FWA alerts by type
loops=1 rows=6 latency=12.56s safe=True
context_precision=0.80 faithfulness=1.00 answer_relevance=1.00

Execution Success Rate (ESR) : 100.0% (6/6)

Self-Healing Efficiency (avg loops): 1.00 (target < 1.5)

False Empty Rate : 0.0% (0/6)

SQL Guardrail Violation Rate : 0.0% (must be 0%)

RAG QUALITY (0..1, LLM-judged)

Context Precision : 0.97

Faithfulness : 1.00

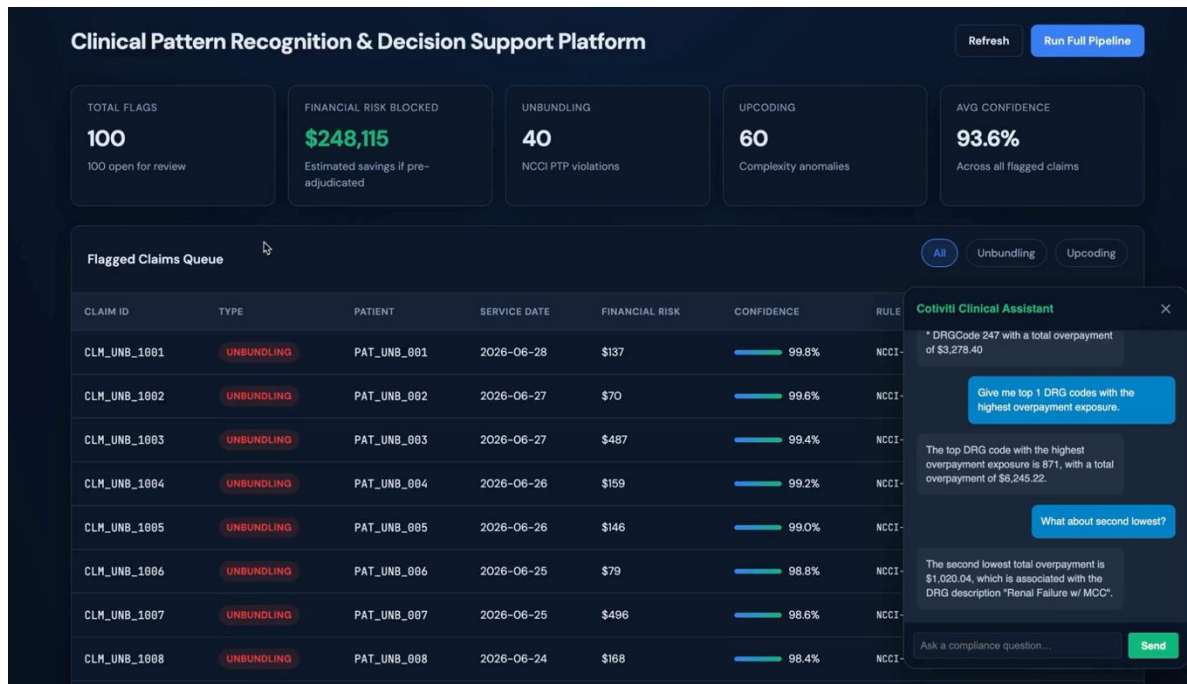
Answer Relevance : 1.00

PERFORMANCE

Avg total latency : 5.21s (TTFT not measured — no streaming)

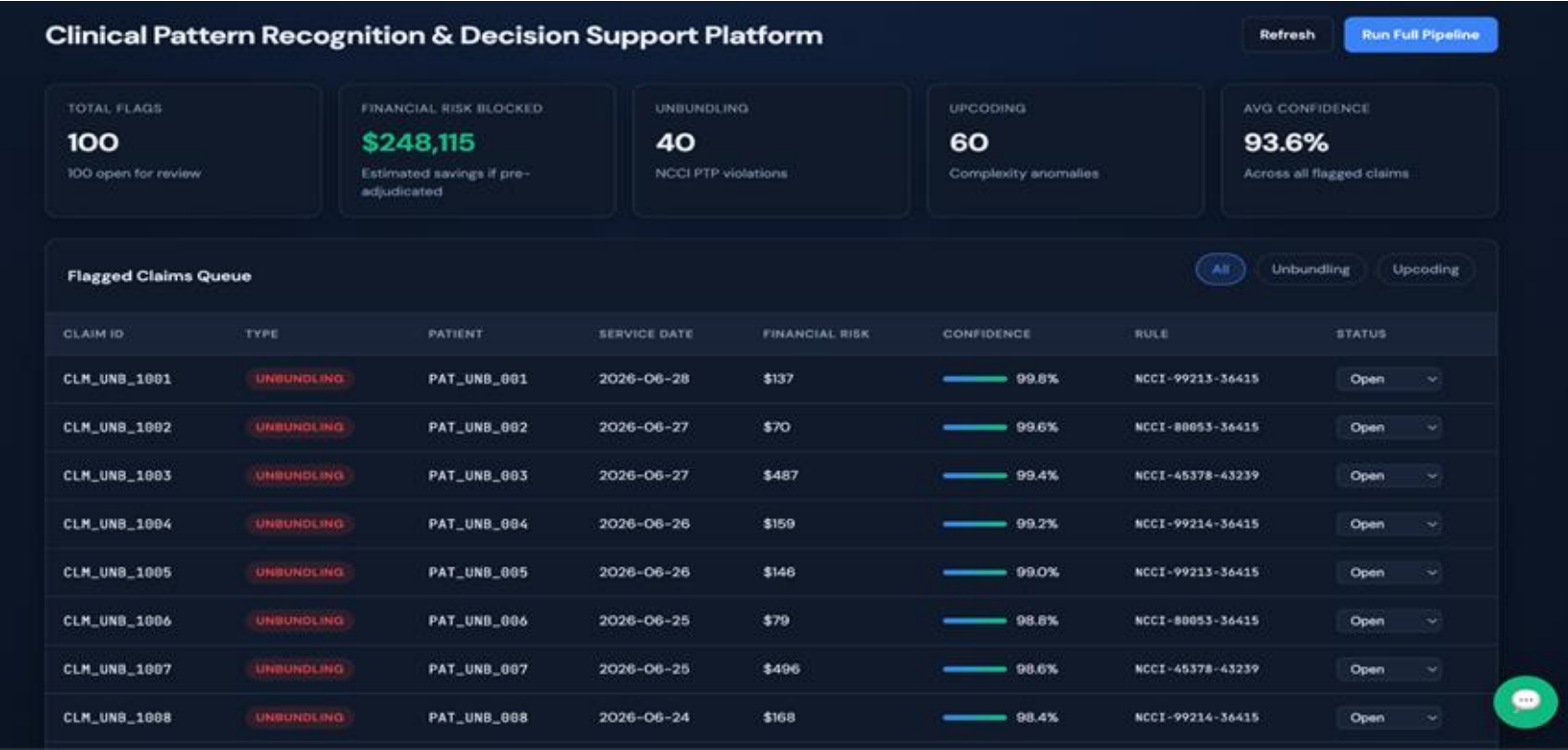
DEMONSTRATION

Platform Demonstration



Web Application Demonstration: ([link](#))

Clinical Pattern Recognition Dashboard



Flagged Claims listing: Unbundling Detection in Action

Flagged Claims Queue							
CLAIM ID	TYPE	PATIENT	SERVICE DATE	FINANCIAL RISK	CONFIDENCE	RULE	STATUS
CLM_UNB_1001	UNBUNDLING	PAT_UNB_001	2026-06-28	\$137	99.8%	NCCI-99213-36415	Open
CLM_UNB_1002	UNBUNDLING	PAT_UNB_002	2026-06-27	\$70	99.6%	NCCI-80853-36415	Open
CLM_UNB_1003	UNBUNDLING	PAT_UNB_003	2026-06-27	\$487	99.4%	NCCI-45378-43239	Open
CLM_UNB_1004	UNBUNDLING	PAT_UNB_004	2026-06-26	\$159	99.2%	NCCI-99214-36415	Open
CLM_UNB_1005	UNBUNDLING	PAT_UNB_005	2026-06-26	\$146	99.0%	NCCI-99213-36415	Open
CLM_UNB_1006	UNBUNDLING	PAT_UNB_006	2026-06-25	\$79	98.8%	NCCI-80853-36415	Open
CLM_UNB_1007	UNBUNDLING	PAT_UNB_007	2026-06-25	\$496	98.6%	NCCI-45378-43239	Open
CLM_UNB_1008	UNBUNDLING	PAT_UNB_008	2026-06-24	\$168	98.4%	NCCI-99214-36415	Open
CLM_UNB_1009	UNBUNDLING	PAT_UNB_009	2026-06-24	\$155	98.2%	NCCI-99213-36415	Open

NCCI Edit Match

Auto-matched to specific CMS edit pair codes

Overpayment Estimate

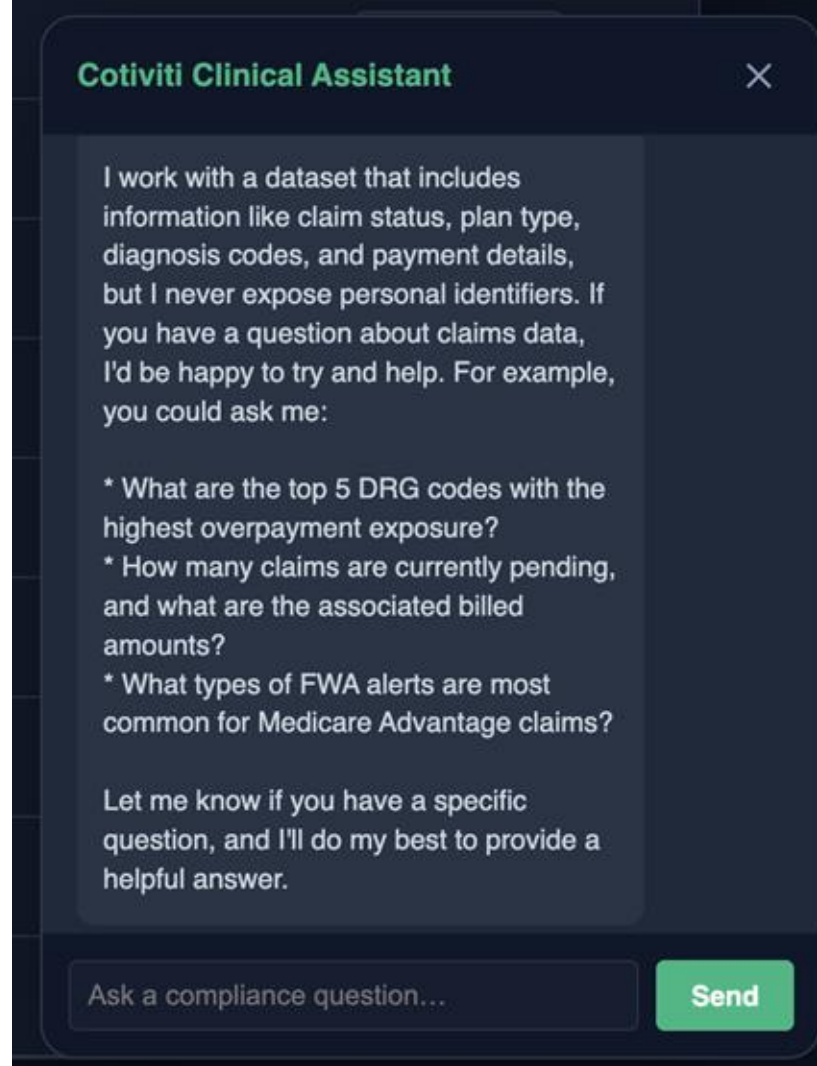
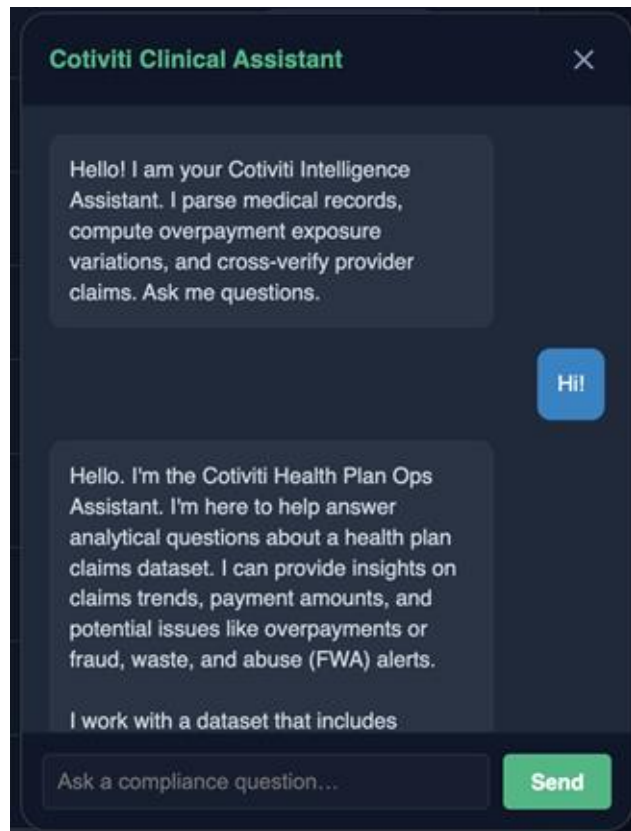
Calculated delta between billed and allowable amounts

Flagged Claims listing: Upcoding Detection in Action

Flagged Claims Queue							
CLAIM ID	TYPE	PATIENT	SERVICE DATE	FINANCIAL RISK	CONFIDENCE	RULE	STATUS
CLM_UPC_2001	UPCODING	PAT_UPC_001	2026-06-28	\$3,669	89.2%	UPCODE-E&M	Open ▾
CLM_UPC_2002	UPCODING	PAT_UPC_002	2026-06-28	\$3,684	89.1%	UPCODE-E&M	Open ▾
CLM_UPC_2003	UPCODING	PAT_UPC_003	2026-06-27	\$3,698	89.0%	UPCODE-E&M	Open ▾
CLM_UPC_2004	UPCODING	PAT_UPC_004	2026-06-27	\$3,713	88.9%	UPCODE-E&M	Open ▾
CLM_UPC_2005	UPCODING	PAT_UPC_005	2026-06-27	\$3,728	88.8%	UPCODE-E&M	Open ▾
CLM_UPC_2006	UPCODING	PAT_UPC_006	2026-06-26	\$3,743	88.7%	UPCODE-E&M	Open ▾
CLM_UPC_2007	UPCODING	PAT_UPC_007	2026-06-26	\$3,757	88.6%	UPCODE-E&M	Open ▾
CLM_UPC_2008	UPCODING	PAT_UPC_008	2026-06-26	\$3,772	88.5%	UPCODE-E&M	Open ▾

DEMONSTRATION

Chat Interface



Chat Interface

Generated SQL for Validation

Interpreted Context Pattern: *What is the second lowest DRG code based on total overpayment amount?*

Extracted Context Token Signatures: DRG code , total overpayment amount , second lowest

Target Columns Retrieved from ChromaDB Index: OverpaymentAmt , DRGDescription , OverpaymentFlag , DRGCode , BilledAmount , PaidAmount , AllowedAmount , PrimaryDx

```
SELECT "DRGCode", SUM("OverpaymentAmt") AS "TotalOverpayment"
FROM claims
GROUP BY "DRGCode"
ORDER BY "TotalOverpayment" ASC
LIMIT 1 OFFSET 1
```

Refinement Optimization Loop Counts: 1

Total Query Pipeline Latency: 2.04s

Cotiviti Clinical Assistant



* DRGCode 247 with a total overpayment of \$3,278.40

Give me top 1 DRG codes with the highest overpayment exposure.

The top DRG code with the highest overpayment exposure is 871, with a total overpayment of \$6,245.22.

What about second lowest?

The second lowest total overpayment is \$1,020.04, which is associated with the DRG description "Renal Failure w/ MCC".

Ask a compliance question...

Send

DEMONSTRATION

SQL Generation: Chat Interface

Generated SQL for Validation

Extracted Context Token Signatures: TOP , 5 , DRG codes

Target Columns Retrieved from ChromaDB Index: DRGDescription , DRGCode , PrimaryDx , FWTtype , PlanType , BilledAmount , ProviderNPI , OverpaymentAmt

```
SELECT "DRGCode", SUM("OverpaymentAmt") AS "TotalOverpayment"
FROM claims
GROUP BY "DRGCode"
ORDER BY "TotalOverpayment" DESC
LIMIT 5
```

Refinement Optimization Loop Counts: 1

Total Query Pipeline Latency: 2.05s

Query claim fields, active overpayment instances, or anomalous DRG codes...

↑

Clear Conversation Memory Workspace

Cotiviti Clinical Assistant

variations, and cross-verify provider claims. Ask me questions.

Give me TOP 5 DRG codes.

The top 5 DRG codes by total overpayment amount are:

- * DRGCode 871 with a total overpayment of \$6,245.22
- * DRGCode 291 with a total overpayment of \$5,156.40
- * DRGCode 378 with a total overpayment of \$3,788.62
- * DRGCode 194 with a total overpayment of \$3,445.53
- * DRGCode 247 with a total overpayment of \$3,278.40

Ask a compliance question...

Send

Conclusion

Possible To:

- Replace slow, manual, rule-based screening with a fast, scalable automated engine.
- Catch statistical anomalies via. Deterministic Gate and Probabilistic Gate.
- Instantly flag fraud schemes before payments are distributed.
- Achieve a 100% Execution Success Rate taking few seconds to response back.
- Fully redact SSNs and mask emails and phone numbers for data privacy.
- Reduce Auditor Time.

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THANK YOU

